Math Olympiad Trainer Academy

AY 25/26 — Cycle II | Exercise Collection

Part I — Invariants and Monovariants

Exercise 1

A pizza is cut in 6 equal pieces and numbers $1, 0, 1, 0, 0, 0$ are written in it following clock-wise direction. In one move we can pick two adjacent parts and increase their numbers by 1. Is it possible to make all numbers inside it equal?

Exercise 2

On a 15×15 table 15 rooks (chess pieces) are placed such that no two of them attack each other (i.e. they are all in different rows and different columns). In one moment each one of them makes an L-jump like a knight in chess would make (1 step in one and 2 steps in other direction). Prove that after this there must be two rooks attacking each other.

Exercise 3

A 3×3 table has been filled with numbers $0, 1, \dots, 8$ in the usual way. In each step pick two adjacent cells and if the numbers in them are x, y and $x \le y$, replace them with $0, y-x$ respectively. Can we obtain a table full of zeroes using these steps?

Exercise 4

If $2n$ points are given in a plane n blue ones and n red ones: show that it is possible to arrange them in pairs of points $A_1, B_1, A_2, B_2, \dots, A_n, B_n$ such that A_i's are red, B_j's are blue and the closed lines $A_1B_1, A_2B_2, \dots, A_nB_n$ never intersect.

Exercise 5

On a square table 10×10, 9 squares are “infected”. We do not know which squares are “infected” initially. Every day after the initial one, new squares get “infected” — namely these ones that share at least two sides with already “infected” ones. Can entire table become infected?

Exercise 6

Initially, the numbers $1, 2, \dots, 2010$ are written on the table. In one step it is allowed to replace numbers a, b with the number $ab+a+b$. Can we ever the number N written on the table if:

1. $N = 2^{2011} - 1$;
2. $N = 2^{2011}$?

Exercise 7

Initially there are n 1's written on a board. In one step numbers x, y are picked and erased and $\frac{x+y}{4}$ is written on the board. After $n-1$ steps only one number is left on the board. Show that the number that is left is not smaller than $\frac{1}{n}$.

Part II — Combinatorial Games

Exercise 8

Ana and Bob have 10 coins on the table. In one move a player can remove 1 or 2 coins from the table. The winner is the one who takes the last coin. Who has the winning strategy?

Exercise 9

Assume that Ana and Bob are playing a game where they have a pile of n stones and in one move the player that makes the move is allowed to pick one or two stones from the pile. If Ana starts first, who has the winning strategy? We assume that the player that cannot make a move — i.e. has an empty table — lost.

Exercise 10

Assume Ana and Bob play the same game, but this time they are allowed to pick $1, 2, 3$ or 4 stones from the pile. Again, they start with n stones. Who has the winning strategy? Again, we assume that the player that cannot make a move — i.e. has an empty table — lost.

Exercise 11

Now start with n coins. On each move a player removes 2 or 3 coins. Who has the winning strategy, in terms of n?

Exercise 12

Use induction on k to prove that $5k$ and $5k+1$ are losing positions, while $5k+2$, $5k+3$, and $5k+4$ are winning positions, in the game where a player removes 2 or 3 coins on each move.

Exercise 13

Start with n stones. On each move a player removes $2, 3$, or 4 stones. Classify the losing positions.

Exercise 14

Two piles. In one move, choose a pile and remove 1 or 2 coins from that pile. Starting from $(5, 4)$, who wins?

Exercise 15

We start with $(5, 4)$ but are allowed to take any number we want from one pile. Who wins?

Exercise 16

For a position with two piles containing (x_1, x_2) coins, when does the first player win if a player may remove any positive number of coins from one pile on each move?

Exercise 17

Assume that the rules hold:

- Players play game with m piles of coins each of which initially have $x_1, x_2, \ldots, x_m$ coins respectively.
- In one move, the player that is playing can pick a pile, say the pile k, and then remove from that pile any number of coins that is available (any number c of coins with $1 \le c \le x_k$).
- A player that wins is the one that picks the last coin — i.e. no coins available after he plays.

Which player has a winning strategy in this game, i.e. what needs to hold such that the first player/second player has the winning strategy?

Exercise 18

Compute $13 \oplus 10$.

Exercise 19

Compute $x \oplus y$ for:

1. $x = 22,\ y = 12$;
2. $x = 18,\ y = 9$;
3. $x = 27,\ y = 14$.

Exercise 20 — Exercise 6.1

Compute the nim-sum and decide whether the position is winning or losing:

$$(7,7,0),(2,4,5),(10,12,6),(1,2,3,4).$$

If it is winning, find one move to a losing position.

Exercise 21 — Exercise 6.2

In the subtraction game “remove 2 or 3”, explain (in one paragraph) why it is hard to guess the strategy by inspection but easy to see it from the W/L table.

Exercise 22 — Exercise 6.3

Design your own impartial game with a small state space (something you can draw as a graph of positions). Label each position W or L using Proposition 2.1, and describe the winning strategy.

Exercise 23 — Exercise 1.1: Warm-up computations

Compute the W/L classification for:

1. $\mathrm{Sub}(\{1,3\})$ up to $n = 15$ and look for a pattern.
2. $\mathrm{Sub}(\{1,4\})$ up to $n = 20$ and describe the losing positions.

Exercise 24 — Exercise 6.1: Small-board analysis

Compute W/L status (and a winning move when it exists) for Chomp on:

1. 2×2;
2. 2×3;
3. 3×3;

by listing all distinct shapes reachable (up to symmetry) and applying W/L recursion.

Exercise 25 — Exercise 11.1: From W/L to strategy

In any finite impartial game under normal play, prove:

1. From any winning position, there exists at least one move to a losing position.
2. If you always move from winning to losing when possible, you will win.

Exercise 26 — Exercise 11.2: Grundy numbers for subtraction games

Let $S \subseteq \mathbb{N}$ be finite. Define $g(n)$ for $\mathrm{Sub}(S)$ by:

$$g(0)=0, g(n)=mex\{g(n-s): s \in S, s \leq n\}.$$

Compute $g(n)$ for $S=\{1,2\}$ up to $n=12$ and compare to the W/L classification.

Exercise 27 — Exercise 11.3: Nim: prove the winning-move formula

Let $s=h_1 \oplus \cdots \oplus h_m \neq 0$ and let b be the most significant 1-bit of s. Prove that there exists a heap h_i with a 1 in bit b and that $h_i'=h_i \oplus s$ satisfies $h_i'<h_i$ and makes the new nim-sum zero.

Exercise 28 — Exercise 11.4: Wythoff Nim small cases

Compute the losing positions of Wythoff Nim with both heaps at most 10 by hand or by a short program. Compare with the pairs $(\lfloor k\varphi \rfloor, \lfloor k\varphi^2 \rfloor)$ for $k=0,1,\ldots$.

Exercise 29 — Project

Construct a NIM-like game and decide which positions are W and which ones are L.

Exercise 30 — Project

Design your own game (it can be very simple) and then derive some laws and strategies that hold inside it.

Part III — Cellular Automata

Exercise 31 — Research-style investigation

Let Q_n be the configuration where every cell in an $n \times n$ square is alive at time 0. What happens to Q_n as time evolves? Does it die out? Stabilize? Produce oscillators? How does the outcome depend on n?

Exercise 32 — Exercise 9.1: Manual Life computation (do before coding)

Work on graph paper. Use the Life rules exactly.

1. Start with a 3×3 filled square. Compute generations $1, 2, 3$.
2. Start with a 4×4 filled square. Compute generation 1 exactly and describe the pattern.
3. In each case, record which births occur outside the original square and why.

Exercise 33 — Example 10.1: A rule change you can explore

Consider B3/S123 (a live cell survives with $1, 2, 3$ neighbors). Predict qualitatively what might happen:

1. Loneliness is less deadly, so small sparse patterns may persist more.
2. Crowding may still destroy dense regions if survival does not include large neighbor counts.

Then test on the same initial conditions you used for Life (e.g., $n \times n$ blocks) and compare outcomes.

Exercise 34 — Exercise 11.5: Life on a torus vs Life on a plane

Simulate the same finite pattern (for example, a glider) under:

1. fixed dead boundary;
2. torus wrap-around;

on an $M \times N$ grid where M, N are small. Describe precisely how and when the behaviors diverge.